《计算机网络》课程作业（9）

（3月29日布置；4月7日课前手写提交）

1. Figure 9-1 shows how addresses can be assigned to computers that connect to a packet switch. Suppose the hardware for one of the interfaces on a switch fails, and a network administrator moves a computer's connection to an unused interface. Will the new configuration work correctly? Why or why not?

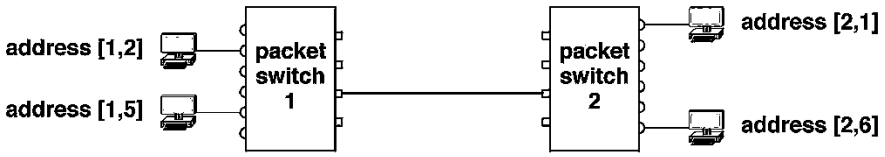


Figure 9-1 Example of an address hierarchy where each address identifies a packet switch and a computer attached to the switch.

2. Default routing helps simplify routing tables. For example, suppose a WAN consists of two packet switches. Each switch can have a routing table entry for each local address (i.e., the address of each computer that attaches to the switch) plus a default route that points to the other switch. Under what circumstances does the scheme fail?

3. When a packet switch receives a distance-vector message from a neighbor, will the switch's forwarding table always change? Explain.

4. 以图 9-2 为例，以最少跳数作为最短路径，给出下面图形中所有交换设备的路由表。（提示，路由表的格式参考图 9-3。）

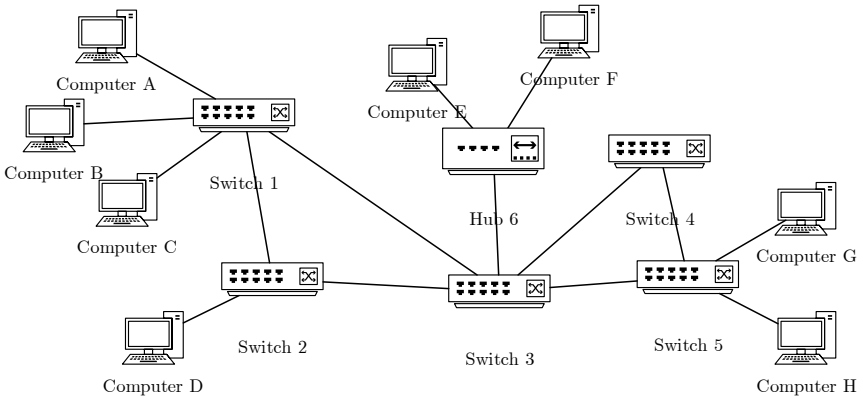


Figure 9-2 A network architecture.

Switch 1	
to reach	**next hop**
1	-
*	(1, 3)

Figure 9-3 An example of forwarding table. (NOT the solution for Fig. 9.2)

5. 图 9-2 中，Computer B 发一个包给 Computer E，其中，Hub 6 连接在 Switcher 3 上的第 2 个端口，如何对 Computer E、Computer F 分别编址？交换机应不知道集线器 Hub 6 的存在，但 Computer E 和 F 确实需要编址，在转发时如何区分 E 和 F？

   （提示：通过查阅资料和讨论，正确理解交换机和集线器的异同。）

6. 网络协议为什么要分层？ISO-OSI 参考模型包括哪些层次，各有什么作用？不同层次数据包结构遵循何种形式设计，以使得各层次可以发挥其作用，又能够不干扰其他层次的功能？

7. 理解 TCP/IP 协议的层次结构和各层功能，比较与 ISO/OSI 参考模型的异同。TCP/IP 网络体系结构为什么要保证网络层的协议一致？